

OBERGURGL, Austria

WMO: 111270

Lat: 46.8668N

Lon: 11.0246E

Elev: 1939

StdP: 80.10

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.0	(c) -14.8	(d) -22.4	(e) 0.6	(f) -14.9	(g) -20.7	(h) 0.8	(i) -12.8	(j) 11.2	(k) -8.1	(l) 9.8	(m) -7.9	(n) 1.4	(o) 210	(p) 0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	21.8	13.0	20.2	12.3	18.7	11.5	13.7	19.9	12.8	18.8	12.0	17.6	2.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	10.6	15.1	10.3	9.9	14.5	9.6	9.4	13.9	45.0	20.1	42.4	19.0	40.0	17.6	18.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.3	5.2	-19.1	24.4	2.9	1.4	-21.2	25.4	-22.9	26.2	-24.5	27.0	-26.6	28.1		
			-19.7	15.2	2.7	1.3	-21.7	16.1	-23.2	16.9	-24.7	17.6	-26.7	18.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-5.7	-4.8	-1.7	2.1	6.0	10.5	11.9	11.7	7.8	4.7	-0.2	-3.5
	DBStd	7.44	4.41	4.88	4.53	3.79	3.88	3.97	3.48	3.45	3.59	4.33	4.82	4.53
	HDD10.0	2700	487	413	364	238	133	43	23	23	82	167	307	418
	HDD18.3	5498	746	647	622	488	382	237	200	206	316	422	557	677
	CDD10.0	244	0	0	0	0	9	56	82	77	17	3	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	12	0	0	0	0	0	6	3	3	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	1.7	1.9	1.8	1.7	1.7	1.9	1.8	1.8	1.7	1.6	1.6	1.6	1.7
(15) Precipitation	PrecAvg	938	46	47	53	58	80	116	119	118	81	85	83	54
	PrecMax	1231	95	126	134	104	147	183	175	229	135	192	277	109
	PrecMin	769	12	12	8	8	19	58	78	45	30	4	8	4
	PrecStd	125	24	29	27	24	37	27	26	37	27	46	56	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	7.2	9.8	13.8	19.1	24.0	23.7	23.7	19.1	16.4	11.0	6.0
		MCWB	1.2	1.0	2.7	6.6	10.3	12.3	13.7	13.7	10.8	8.3	5.1	1.2
	2%	DB	3.3	4.7	7.7	11.4	16.6	20.9	21.7	21.5	17.2	13.8	8.3	4.2
		MCWB	-0.3	0.2	2.0	5.0	9.1	12.1	13.3	13.0	10.1	7.2	3.4	0.4
	5%	DB	1.7	2.7	5.8	9.4	14.5	19.2	20.1	19.7	15.2	11.7	6.6	2.9
		MCWB	-1.4	-0.9	0.9	4.1	7.9	11.6	12.5	12.3	9.4	6.2	2.7	-0.2
	10%	DB	0.4	1.2	4.1	7.7	12.5	17.3	18.4	17.9	13.4	9.9	5.2	1.6
		MCWB	-2.2	-2.0	0.2	3.1	6.8	10.4	11.6	11.5	8.6	5.6	1.9	-1.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.1	3.6	7.1	11.0	13.6	15.4	14.7	12.5	9.4	5.9	2.2
		MCDB	5.1	5.7	8.4	13.0	17.8	20.4	20.8	21.1	16.8	14.0	9.9	4.4
	2%	WB	0.2	0.6	2.4	5.6	9.6	12.7	14.0	13.6	11.0	8.0	4.2	1.3
		MCDB	2.2	3.7	6.5	10.4	15.5	19.4	20.2	19.8	15.2	12.0	7.0	3.5
	5%	WB	-0.8	-0.6	1.5	4.4	8.4	11.8	13.0	12.7	10.0	7.0	3.2	0.2
		MCDB	1.3	2.4	5.0	8.7	13.5	18.4	18.9	18.4	14.0	10.4	5.7	2.3
	10%	WB	-2.0	-1.7	0.5	3.4	7.3	10.9	12.0	11.8	9.1	6.1	2.3	-1.0
		MCDB	0.1	1.0	3.6	7.2	11.7	16.5	17.3	16.9	12.7	9.4	4.8	1.1
(35) Mean Daily Temperature Range	MDBR		5.8	7.2	8.1	7.9	8.5	9.1	9.3	9.1	8.2	7.7	5.9	5.5
	5% DB	MCDBR	6.9	8.5	9.4	9.6	11.3	11.9	12.0	11.5	10.7	9.7	6.8	6.3
		MCWBR	5.2	5.9	6.2	5.7	5.9	5.5	5.4	5.5	5.7	5.6	4.4	4.4
	5% WB	MCDBR	6.3	7.8	8.5	8.9	10.6	11.0	11.0	10.8	9.4	8.8	6.0	6.0
		MCWBR	5.2	5.6	5.7	5.3	5.8	5.3	5.4	5.5	5.2	5.3	4.2	4.8
(40) Clear-Sky Solar Irradiance	taub		0.224	0.246	0.284	0.324	0.339	0.346	0.343	0.337	0.309	0.288	0.255	0.225
	taud		2.384	2.333	2.309	2.336	2.373	2.389	2.392	2.428	2.497	2.532	2.529	2.445
	Ebn at Noon		904	943	945	931	925	917	914	908	910	884	858	865
	Edh at Noon		67	89	108	116	117	116	114	105	89	74	60	57
(44) All-Sky Solar Radiation	RadAvg		1.53	2.47	3.65	4.44	4.87	5.53	5.55	4.87	3.73	2.52	1.51	1.26
	RadStd		0.17	0.29	0.43	0.60	0.54	0.47	0.45	0.47	0.40	0.31	0.24	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (95 neighbors)		+0.35	N/A	N/A	+0.64	+0.41	+0.37	N/A	N/A	+61

Nomenclature: See separate page